Unit 2 Lecture 8 CMOS Inverters F24

Wednesday, September 11, 2024 12:50 PM

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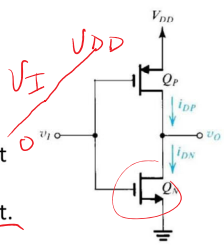
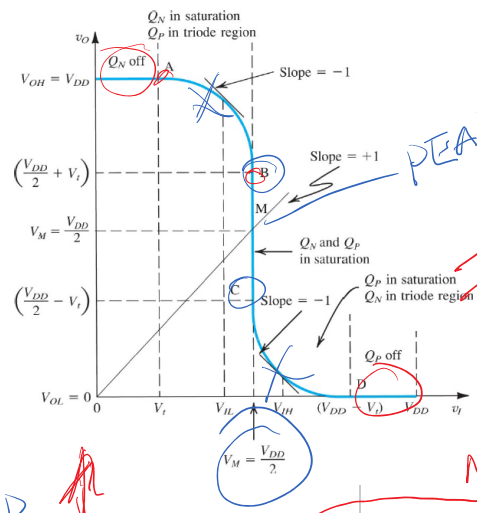
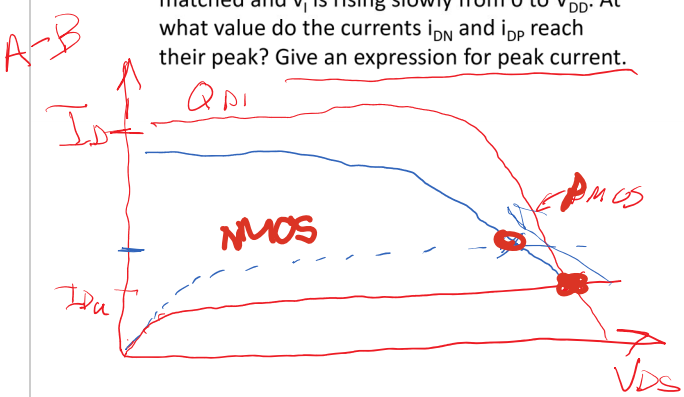
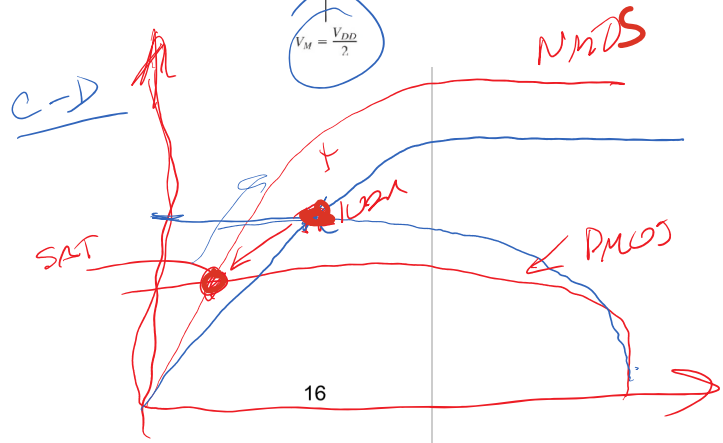
Problem 16.33

For the CMOS inverter shown, Q_N and Q_P are matched and v_I is rising slowly from 0 to V_{DD}. At what value do the currents i_{DN} and i_{DP} reach their peak? Give an expression for peak current.



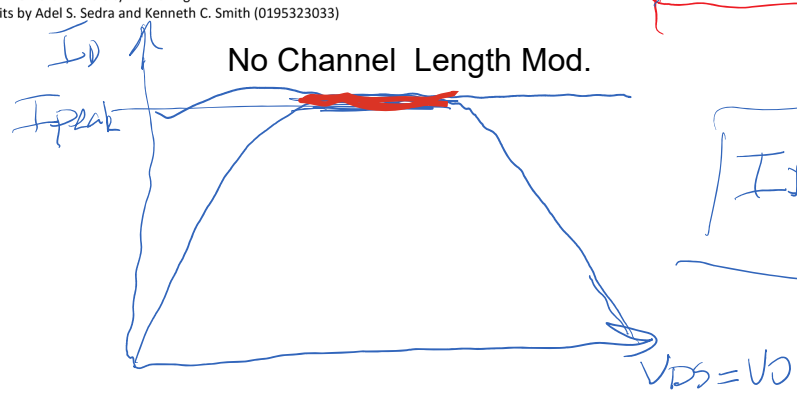






Oxford University Publishing
Microelectronic Circuits by Adel S. Sedra and Kenneth C. Smith (0195323033)

No Channel Length Mod.



$$I_{D_N} = \frac{K}{2}\left(\frac{V_{DD}}{2} - V_T\right)^2$$

V_I

VIL = 1.06V

1.47V = VIM

What is noise margin?

16.33 Consider the CMOS inverter of Fig. 16.22 with QN and QP matched and with the input vI rising slowly from 0 to VDD. At what value of vI does the current flowing through QN and QP reach its peak? Give an expression for the peak current, neglecting λn and λp. For = 540 μA/V2, (W/L)n = 1.5, VDD = 1.0 V, and Vtn = 0.35V, find the value of the peak current.

[Kenneth C. (KC) Smith. Microelectronic Circuits]

16.33 The current reaches its peak at $v_I = V_M = \frac{V_{DD}}{2}$. At this point, both Q_N and Q_P are operating in the saturation region and conducting a current

$$I_{DP} = I_{DN} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n\left(\frac{V_{DD}}{2} - V_t\right)^2$$

$$= \frac{1}{2} \times 540 \times 1.5\left(\frac{1.0}{2} - 0.35\right)^2$$

$$= 9.1\ \mu A$$